EcoSort AI: Real-Photo Update

23 October 2026 | Current application and sixty-photo acceptance evidence

The application now uses a pinned Google SigLIP2 base model locally on the Mac. Object descriptions are matched to the full photograph, then mapped to 17 project categories. A separate City of Sydney / NSW EPA rule module supplies conditional disposal advice. This is a zero-shot object recognizer, not a newly trained 147,303-image classifier.

Panel	Category correct	Category + route correct	Required
1	19/20	19/20 (95%)	17/20
2	18/20	18/20 (90%)	17/20
3	20/20	18/20 (90%)	17/20

All three panels passed on their first recorded evaluation. Total category accuracy is 57/60 (95%); total category-and-conditional-route accuracy is 55/60 (91.7%). No failed image was replaced after predictions. Uncertain outputs count as joint failures, even when the suggested category is correct.

Limits: these are stratified convenience samples from Wikimedia Commons, not randomly sampled Sydney household photos. Sixty examples do not establish universal 80% or 95% real-world accuracy. Foundation-model pretraining membership is unknown. The mixed-items class is not covered. Routes do not verify hidden contents, contamination, damaged batteries, or access to a service.

The six previously observed phone photos improved from the former deployed model's 2/6 to 5/6 category matches. They are development examples, not fresh validation. The blue-green plastic puzzle remains an uncertain incorrect cardboard suggestion. No private phone photo is included in the public repository.

What changed and what did not

Component	Current implementation
Object model	Google SigLIP2 base patch16 224; frozen weights; local MPS execution; individual object prompts instead of a benchmark-trained material head.
Inputs	Whole RGB photo with EXIF orientation corrected, then official model preprocessing; no manual crop or ground-truth item confirmation supplied during acceptance.
Category	Maximum object-text similarity per class. Relative softmax scores are displayed as uncalibrated candidate scores, not accuracy probabilities.
Uncertainty	Review if top two class logits differ by less than 0.5, or highest logit is below -10. Frozen before acceptance prediction.
Disposal	Inferred item guidance is conditional and distinct from a user-confirmed item. Batteries, lamps and paint need special collection; soft plastic and paper towels are not yellow-bin recycling.
Dataset history	167,285 indexed images; 147,303 retained; 117,893 train / 14,698 validation / 14,712 test. These counts describe archived project experiments, not new SigLIP2 training.
Historical benchmark	95.13% validation and 94.85% public test belong to an archived CLIP+DINO candidate. The public test has been reused; do not attribute these numbers to the current application.

Key files: object_model.py, sydney_rules.py, ecosort_runtime.py, active_model.json, trained_objects/deployment_approved.json, acceptance_web/test_manifest.json, acceptance_web/results.csv, and test_runtime_acceptance.py. The English local-run notebook retains explanation/code cell pairs.

Setup: install requirements-lock.txt in a Python 3.12 virtual environment; run python object_model.py once to download the pinned ~1.50 GB model; then run python app.py. Inference remains local. To restore the frozen web panel, run python restore_acceptance.py, then python test_runtime_acceptance.py. Download failures or changed source files must not silently replace an evaluated image.

Every failure is retained

Panel / ID	Ground truth	Actual output	Why joint failure
2 / 0:1	Plastic drink bottle	glass bottle	Manual review
3 / 0:2	Plastic drink bottle	plastic drink bottle	Manual review
1 / 21:0	Aluminium beverage can	plastic packaging	Manual review
3 / 21:2	Aluminium beverage can	empty tin can	Manual review
2 / 15:1	Soft plastic shopping bag	sheet of printed paper	Wrong category and route

The printed soft-plastic bag was confidently mistaken for paper and received an incorrect recycling route. This is a remaining safety-relevant failure, not an item to hide from the report. The application must not be used for unattended hazardous-waste decisions.

Protocol: twenty predefined object slots, with three different photographs per slot. Visual review excluded drawings, unrelated search results, ambiguous material, and repeated views before model prediction. SHA-256, resized-pixel hashes and perceptual hashes were checked against the entire project pool and prior panels; no detected duplicates were found. These are heuristic checks, not proof of complete independence.

Official rule source: City of Sydney household recycling. Additional food, garden, doorstep and NSW EPA links are embedded in the runtime rule module. Google model source: SigLIP2 official model card.

Panel 1: all twenty outcomes

Source-page links are attached to each photo ID. Complete route outputs, hashes, author and licence metadata are in the accompanying JSON/CSV files.

Photo ID	Reference object	Predicted object	Joint result
0:0	Plastic drink bottle	plastic drink bottle	Correct
1:0	Rigid plastic food packaging	black plastic food container	Correct
2:0	Glass bottle	glass bottle	Correct
3:0	Glass jar	glass jar	Correct
21:0	Aluminium beverage can	plastic packaging	FAILED
5:5	Metal food tin	steel food tin	Correct
6:0	Cardboard shipping box	corrugated cardboard box	Correct
7:1	Newspaper	newspaper	Correct
8:3	Household battery	household battery	Correct
9:3	Computer keyboard	computer keyboard	Correct
10:2	Fabric shirt	cotton clothing	Correct
11:0	Porcelain mug	ceramic mug	Correct
12:3	Food scraps / fruit peel	banana peel	Correct
27:0	Fallen leaves	pile of fallen leaves	Correct
14:2	Wooden spoon	wooden utensil	Correct
15:0	Soft plastic shopping bag	plastic bag	Correct
16:0	Paper kitchen towel	roll of paper kitchen towel	Correct
17:1	Compact fluorescent lamp	light bulb	Correct
18:0	Medicine tablets in packaging	pharmaceutical blister pack	Correct
25:0	Paint in cans	paint can	Correct

Photo sources: 0:0 | 1:0 | 2:0 | 3:0 | 21:0 | 5:5 | 6:0 | 7:1 | 8:3 | 9:3 | 10:2 | 11:0 | 12:3 | 27:0 | 14:2 | 15:0 | 16:0 | 17:1 | 18:0 | 25:0

Panel 2: all twenty outcomes

Source-page links are attached to each photo ID. Complete route outputs, hashes, author and licence metadata are in the accompanying JSON/CSV files.

Photo ID	Reference object	Predicted object	Joint result
0:1	Plastic drink bottle	glass bottle	FAILED
1:1	Rigid plastic food packaging	plastic food container	Correct
2:2	Glass bottle	glass bottle	Correct
3:2	Glass jar	glass jar	Correct
21:1	Aluminium beverage can	aluminium beverage can	Correct
29:0	Metal food tin	metal food can	Correct
6:2	Cardboard shipping box	corrugated cardboard box	Correct
7:3	Newspaper	newspaper	Correct
8:4	Household battery	rechargeable battery	Correct
9:4	Computer keyboard	computer keyboard	Correct
10:4	Fabric shirt	cotton clothing	Correct
11:1	Porcelain mug	ceramic mug	Correct
31:1	Food scraps / fruit peel	food scraps	Correct
32:1	Fallen leaves	pile of fallen leaves	Correct
14:3	Wooden spoon	wooden utensil	Correct
15:1	Soft plastic shopping bag	sheet of printed paper	FAILED
16:1	Paper kitchen towel	roll of paper kitchen towel	Correct
17:2	Compact fluorescent lamp	fluorescent light tube	Correct
18:3	Medicine tablets in packaging	pharmaceutical blister pack	Correct
25:1	Paint in cans	paint can	Correct

Photo sources: 0:1 | 1:1 | 2:2 | 3:2 | 21:1 | 29:0 | 6:2 | 7:3 | 8:4 | 9:4 | 10:4 | 11:1 | 31:1 | 32:1 | 14:3 | 15:1 | 16:1 | 17:2 | 18:3 | 25:1

Panel 3: all twenty outcomes

Source-page links are attached to each photo ID. Complete route outputs, hashes, author and licence metadata are in the accompanying JSON/CSV files.

Photo ID	Reference object	Predicted object	Joint result
0:2	Plastic drink bottle	plastic drink bottle	FAILED
1:2	Rigid plastic food packaging	plastic food container	Correct
2:4	Glass bottle	glass bottle	Correct
3:3	Glass jar	glass jar	Correct
21:2	Aluminium beverage can	empty tin can	FAILED
29:1	Metal food tin	metal food can	Correct
6:4	Cardboard shipping box	cardboard packaging	Correct
7:4	Newspaper	magazine	Correct
20:1	Household battery	household battery	Correct
9:5	Computer keyboard	computer keyboard	Correct
10:5	Fabric shirt	cotton clothing	Correct
11:2	Porcelain mug	ceramic mug	Correct
35:0	Food scraps / fruit peel	banana peel	Correct
36:0	Fallen leaves	pile of fallen leaves	Correct
14:5	Wooden spoon	wooden utensil	Correct
15:2	Soft plastic shopping bag	plastic bag	Correct
16:4	Paper kitchen towel	roll of paper kitchen towel	Correct
17:3	Compact fluorescent lamp	light bulb	Correct
18:4	Medicine tablets in packaging	pharmaceutical blister pack	Correct
25:3	Paint in cans	paint can	Correct

Photo sources: 0:2 | 1:2 | 2:4 | 3:3 | 21:2 | 29:1 | 6:4 | 7:4 | 20:1 | 9:5 | 10:5 | 11:2 | 35:0 | 36:0 | 14:5 | 15:2 | 16:4 | 17:3 | 18:4 | 25:3